

Active-Learning-Assisted Phase Boundary Refinement for 1D Altermagnetic FFLO TSC

Auto-generated by `ml_phase.report_builder`

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1 Physical and Computational Motivation

The exact BdG solver is expensive on dense grids. We use active learning to prioritize exact calls near uncertain or boundary-like regions.

2 Warm-Start Dataset

Run ID: `RUN_ID`

Warm-start samples: `N_WARM_START`

Current exact samples: `N_EXACT`

Iterations recorded: `N_ITS`

3 Feature Construction

Raw coordinates:

$$x = [k_B T / t, J_A / t].$$

Regression targets:

$$[\Delta_{\text{opt}}, q_{\text{opt}}, \eta, I_c^+, I_c^-].$$

4 Feature Classification and Phase Labels

Class / label	Definition
normal	$\Delta_{\text{opt}} < \Delta_{\epsilon}$
uniform_SC	$\Delta_{\text{opt}} \geq \Delta_{\epsilon}$ and $ q_{\text{opt}} < q_{\epsilon}$
FFLO	$\Delta_{\text{opt}} \geq \Delta_{\epsilon}$ and $ q_{\text{opt}} \geq q_{\epsilon}$
eta_pos	$\eta > 0$
eta_neg	$\eta < 0$
strong_diode	$ \eta > \eta_{\text{strong}}$

Table 1: Label definitions used in the active-learning loop.

5 ML Surrogate and Classifier Models

- Regression model family: `REG_MODELClassificationmodelfamily : CLS_MODEL`